Percentages

↳ Percent (%) :- Per|cent = /100

↳ the basic defination as per name suggest is '$\frac{\quad}{100}$' means for every '100'.

↳ The important principle/Rule/technique on this % concept is

Ref value/Original/Total/Original/Initial value as → 100%

↳ Most of the problems on '%' becomes easy & quick by using above 'matrix'.

$$\% = \frac{\quad}{100} = \text{for every 100.}$$

↳ Conversion from % to decimal/fraction:-

Ex:- $30\% = \frac{30}{100} = \frac{3}{10} = 0.3$ (fraction, decimal)

Note:- Convert from % to dec/fract, we have to $\frac{\quad}{100}$ the given % value.

↳ Conversion from decimal/fraction to %:-

$\left[0.7 \text{ (or) } \frac{7}{10}\right] \times 100 = 70\%$

Note:- to convert fract/dec to %, multiply with 100.

↳ Lets we will go for small analysis:-

Assume 1500 → 100%

150.0 → 10%

15.00 → 1%

$\frac{1500}{2} = 750 \rightarrow 50\% = \frac{100\%}{2}$

$\frac{750}{2} = 375 \rightarrow 25\% = \frac{100\%}{4}$

$\frac{750}{10} = 5 \times 15 = 75 \rightarrow 5\% = 5 \times 1\% = \frac{50\%}{10}$

$7 \times 15 = 105 \rightarrow 7\% = 7 \times 1\% = 5 \times 1\% + 2 \times 1\%$

87% = ?

87% = 8×10% + 7×1% (or)

= 75% + 12% = 3×25% + 12%

= 100% − 13%

= 8×150 + 7×15

= 1305

Note:- The above funda is helping us w/o any formula.

(17) ↳ Creating formula for % value :-

Ex:- 70% of 2000 $= \frac{70}{100} \times 2000 =$ % value (Number)

% value $= \frac{\text{Given \%}}{100} \times$ Total value.

From this we can find out value % (%)

Given % (or) value % $= \frac{\text{\% value / Given value} \times 100}{\text{Total value}}$

★ ↳ Converted table from fractions to %'s

Like:- $\frac{1}{1} \times 100 = 100\%$. or $100\% = \frac{100}{100} = \frac{1}{1}$

$\frac{1}{1} = 100\% = \frac{2}{2} = \frac{3}{3}$ ....

$\frac{1}{2} = \frac{1}{2} \times 100 = 50\%$, $\frac{2}{2} = 100\%$

$\frac{1}{3} = \frac{1}{3} \times 100 = 33\frac{1}{3} = 33.33\%$, $\frac{2}{3} = 2 \times \frac{1}{3} = 66.66\% / 66\frac{2}{3}$

$\frac{1}{4} = \frac{1}{4} \times 100 = 25\%$, $\frac{2}{4} = 2 \times 25\% = 50\%$, $\frac{3}{4} = 75\%$, $\frac{4}{4} = 100\%$

$\frac{1}{5} = \frac{1}{5} \times 100 = 20\%$, $\frac{2}{5} = 40\%$, $\frac{3}{5} = 60\%$, $\frac{4}{5} = (1 - \frac{1}{5}) = 100\% - 20\% = 80\%$.

$\frac{1}{6} = \frac{1}{6} \times 100 = 16.66\% = 16\frac{2}{3}\%$, $\frac{2}{6} = 33\frac{1}{3}$, $\frac{3}{6} = 50\%$, $\frac{4}{6} = 66\frac{2}{3}\%$, $\frac{5}{6} = 83.33\%$

$\frac{1}{7} = \frac{1}{7} \times 100 = 14.28\% = 14\frac{2}{7}\%$, $\frac{2}{7} = 28.56\%$, ... $\frac{6}{7} = 1 - \frac{1}{7} = 100\% - 14.28\% = 85.72\%$

$\frac{1}{8} = \frac{1}{8} \times 100 = 12.5\%$, $\frac{2}{8} = 25\%$ - - - $\frac{7}{8} = 1 - \frac{1}{8} = 100\% - 12.5\% = 87.5\%$.

$\frac{1}{9} = \frac{1}{9} \times 100 = 11\frac{1}{9} = 11.11\%$, $\frac{2}{9} = 22.22\%$ ..... $\frac{8}{9} = 88.88\%$

$\frac{1}{10} = \frac{1}{10} \times 100 = 10\%$, - - - - $\frac{9}{10} = 90\%$

$\frac{1}{11} = \frac{1}{11} \times 100 = 9\frac{1}{11} = 09.09$, - - - - - $\frac{10}{11} = 90.9\%$.

$\frac{1}{12} = \frac{1}{12} \times 100 = \frac{1}{2} = 8.33\%$, $\frac{2}{12} = 16.66\%$ ..... $\frac{11}{12} = 1 - \frac{1}{12} = 91.66\%$

* In the above table there is relation b/w $\frac{1}{9}$ values & $\frac{1}{11}$ values in such a way that for $\frac{1}{9}$, % values are multiples of '9' before 2 tables decimal for numerator no. of times. the same way to $\frac{1}{11}$, % values are multiples of '9' before 2 tables decimal for numerator no. of times $\left[\frac{3}{11}\right]$ 3 times